```
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import pandas as pd
import seaborn as sns
from sklearn.preprocessing import OrdinalEncoder,StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import
accuracy_score,confusion_matrix,recall_score

df = pd.read_csv("customer_churn_dataset-testing-master.csv")

df
```

	CustomerID	Age	Gender	Tenure	Usage Frequency	Support Calls	\
0	1	22	Female	25	14	4	
1	2	41	Female	28	28	7	
2	3	47	Male	27	10	2	
3	4	35	Male	9	12	5	
4	5	53	Female	58	24	9	
...	...	...	...	...	...	...	
64369	64370	45	Female	33	12	6	
64370	64371	37	Male	6	1	5	
64371	64372	25	Male	39	14	8	
64372	64373	50	Female	18	19	7	
64373	64374	52	Female	45	15	9	

	Payment Delay	Subscription Type	Contract Length	Total Spend	\
0	27	Basic	Monthly	598	
1	13	Standard	Monthly	584	
2	29	Premium	Annual	757	
3	17	Premium	Quarterly	232	
4	2	Standard	Annual	533	
...	...	...	...	...	
64369	21	Basic	Quarterly	947	
64370	22	Standard	Annual	923	
64371	30	Premium	Monthly	327	
64372	22	Standard	Monthly	540	

```
64373              25          Standard         Monthly           696
```

```
       Last Interaction  Churn
0                     9      1
1                    20      0
2                    21      0
3                    18      0
4                    18      0
...                 ...    ...
64369                14      1
64370                 9      1
64371                20      1
64372                13      1
64373                22      1

[64374 rows x 12 columns]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 64374 entries, 0 to 64373
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   CustomerID         64374 non-null  int64
 1   Age                64374 non-null  int64
 2   Gender             64374 non-null  object
 3   Tenure             64374 non-null  int64
 4   Usage Frequency    64374 non-null  int64
 5   Support Calls      64374 non-null  int64
 6   Payment Delay      64374 non-null  int64
 7   Subscription Type  64374 non-null  object
 8   Contract Length    64374 non-null  object
 9   Total Spend        64374 non-null  int64
 10  Last Interaction   64374 non-null  int64
 11  Churn              64374 non-null  int64
dtypes: int64(9), object(3)
memory usage: 5.9+ MB
```

```
df.isnull().sum()
```

```
CustomerID           0
Age                  0
Gender               0
Tenure               0
Usage Frequency      0
Support Calls        0
Payment Delay        0
Subscription Type    0
Contract Length      0
```

```
Total Spend          0
Last Interaction     0
Churn                0
dtype: int64
```

```
df.describe()
```

```
         CustomerID           Age        Tenure  Usage Frequency  \
count  64374.000000  64374.000000  64374.000000     64374.000000
mean   32187.500000     41.970982     31.994827        15.080234
std    18583.317451     13.924911     17.098234         8.816470
min        1.000000     18.000000      1.000000         1.000000
25%    16094.250000     30.000000     18.000000         7.000000
50%    32187.500000     42.000000     33.000000        15.000000
75%    48280.750000     54.000000     47.000000        23.000000
max    64374.000000     65.000000     60.000000        30.000000

       Support Calls  Payment Delay   Total Spend  Last Interaction  \
count   64374.000000   64374.000000  64374.000000      64374.000000
mean        5.400690      17.133952    541.023379         15.498850
std         3.114005       8.852211    260.874809          8.638436
min         0.000000       0.000000    100.000000          1.000000
25%         3.000000      10.000000    313.000000          8.000000
50%         6.000000      19.000000    534.000000         15.000000
75%         8.000000      25.000000    768.000000         23.000000
max        10.000000      30.000000   1000.000000         30.000000

              Churn
count  64374.000000
mean       0.473685
std        0.499311
min        0.000000
25%        0.000000
50%        0.000000
75%        1.000000
max        1.000000
```

```
df.shape
```

```
(64374, 12)
```

```
df
```

```
       CustomerID  Age  Gender  Tenure  Usage Frequency  Support Calls
\
0               1   22  Female      25               14              4

1               2   41  Female      28               28              7

2               3   47    Male      27               10              2
```

```
3               4   35    Male       9               12               5

4               5   53  Female      58               24               9

...           ...  ...     ...     ...              ...             ...

64369       64370   45  Female      33               12               6

64370       64371   37    Male       6                1               5

64371       64372   25    Male      39               14               8

64372       64373   50  Female      18               19               7

64373       64374   52  Female      45               15               9


       Payment Delay Subscription Type Contract Length  Total Spend  \
0                 27             Basic         Monthly          598
1                 13          Standard         Monthly          584
2                 29           Premium          Annual          757
3                 17           Premium       Quarterly          232
4                  2          Standard          Annual          533
...              ...               ...             ...          ...
64369             21             Basic       Quarterly          947
64370             22          Standard          Annual          923
64371             30           Premium         Monthly          327
64372             22          Standard         Monthly          540
64373             25          Standard         Monthly          696

       Last Interaction  Churn
0                     9      1
1                    20      0
2                    21      0
3                    18      0
4                    18      0
...                 ...    ...
64369                14      1
64370                 9      1
64371                20      1
64372                13      1
64373                22      1

[64374 rows x 12 columns]
```

```
df.duplicated().sum()
```

```
0
```

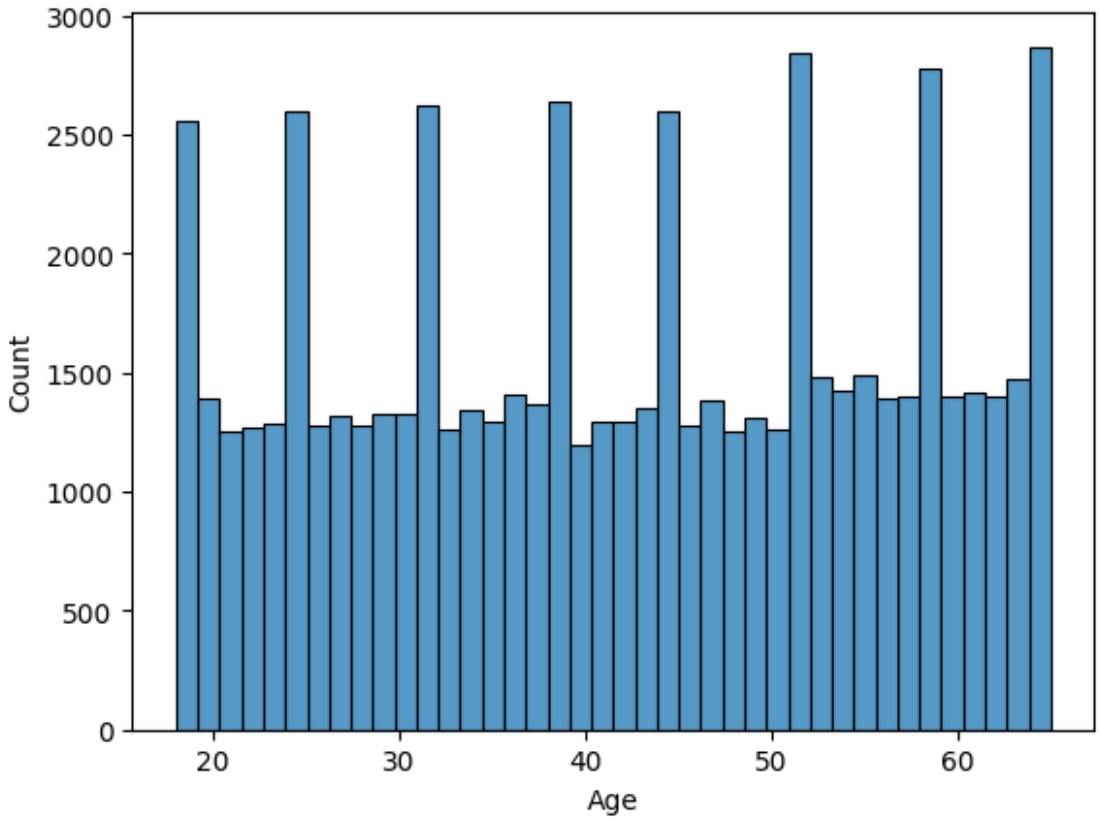
```
df["Churn"].value_counts()
```

```
0    33881
1    30493
Name: Churn, dtype: int64
```

```
df.drop("CustomerID",inplace=True,axis=1)
```

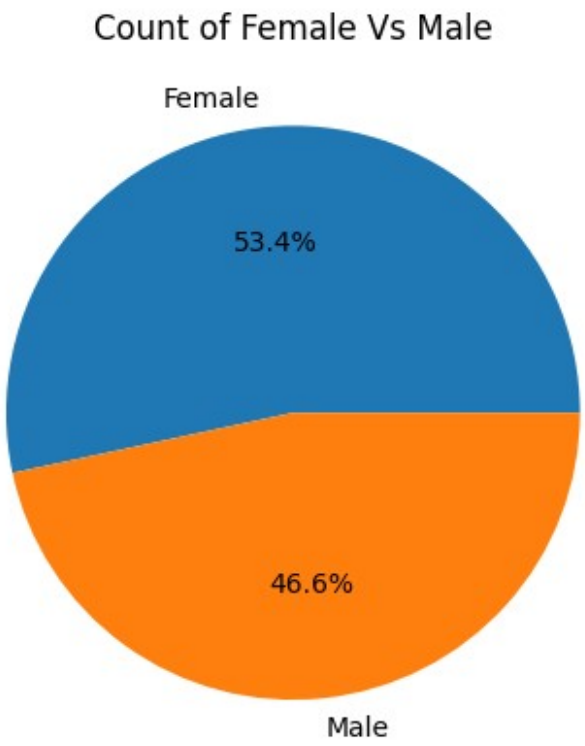
```
sns.histplot(data=df,x="Age")
```

```
<AxesSubplot:xlabel='Age', ylabel='Count'>
```




```
gender_counts = df["Gender"].value_counts()

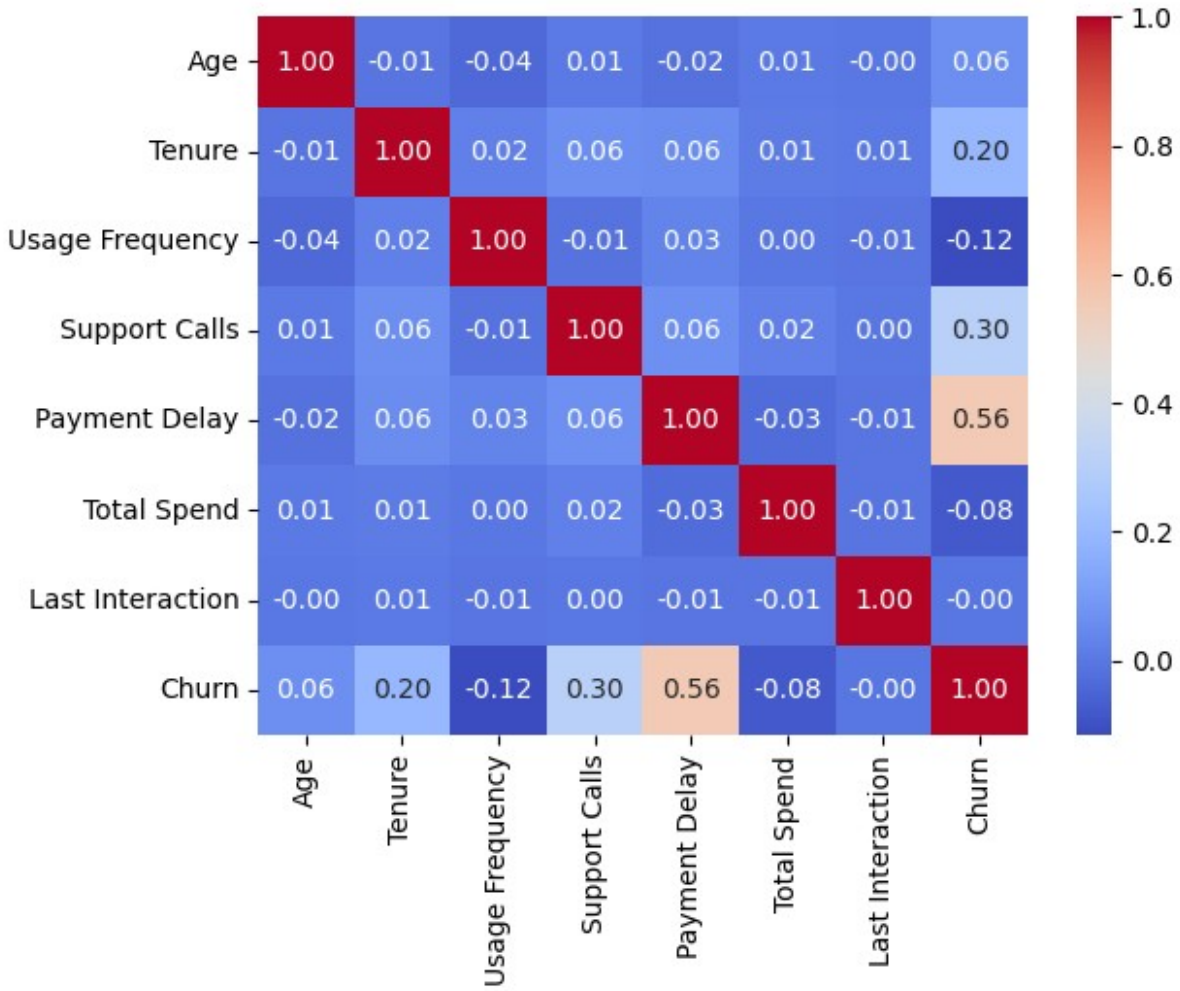
plt.pie(gender_counts,labels=gender_counts.index,autopct="%1.1f%%")
plt.title("Count of Female Vs Male")
plt.show()
```




```
corr_matrix = df.corr()

sns.heatmap(corr_matrix,annot=True,cmap="coolwarm",fmt=".2f")
```

```
<AxesSubplot:>
```




```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 64374 entries, 0 to 64373
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Age                64374 non-null  int64
 1   Gender             64374 non-null  object
 2   Tenure             64374 non-null  int64
 3   Usage Frequency    64374 non-null  int64
 4   Support Calls      64374 non-null  int64
 5   Payment Delay      64374 non-null  int64
 6   Subscription Type  64374 non-null  object
 7   Contract Length    64374 non-null  object
 8   Total Spend        64374 non-null  int64
 9   Last Interaction   64374 non-null  int64
 10  Churn              64374 non-null  int64
```

```
dtypes: int64(8), object(3)
memory usage: 5.4+ MB
```

```
df.head()
```

```
   Age  Gender  Tenure  Usage Frequency  Support Calls  Payment Delay
\
0   22  Female      25               14              4             27

1   41  Female      28               28              7             13

2   47    Male      27               10              2             29

3   35    Male       9               12              5             17

4   53  Female      58               24              9              2


  Subscription Type Contract Length  Total Spend  Last Interaction
Churn
0             Basic         Monthly          598                 9
1
1          Standard         Monthly          584                20
0
2           Premium          Annual          757                21
0
3           Premium       Quarterly          232                18
0
4          Standard          Annual          533                18
0
```

```
df["Gender"] = df["Gender"].map({"Male":0,"Female":1}).astype("int")
```

```
df
```

```
       Age  Gender  Tenure  Usage Frequency  Support Calls  Payment
Delay  \
0       22       1      25               14              4
27
1       41       1      28               28              7
13
2       47       0      27               10              2
29
3       35       0       9               12              5
17
4       53       1      58               24              9
2
...    ...     ...     ...              ...            ...
...
64369   45       1      33               12              6
21
```

```
64370   37       0       6                1              5
22
64371   25       0      39               14              8
30
64372   50       1      18               19              7
22
64373   52       1      45               15              9
25

      Subscription Type Contract Length  Total Spend  Last Interaction
Churn
0                 Basic         Monthly          598                 9
1
1              Standard         Monthly          584                20
0
2               Premium          Annual          757                21
0
3               Premium       Quarterly          232                18
0
4              Standard          Annual          533                18
0
...                 ...             ...          ...               ...
...
64369             Basic       Quarterly          947                14
1
64370          Standard          Annual          923                 9
1
64371           Premium         Monthly          327                20
1
64372          Standard         Monthly          540                13
1
64373          Standard         Monthly          696                22
1

[64374 rows x 11 columns]
```

```
df["Subscription Type"].unique()
```

```
array(['Basic', 'Standard', 'Premium'], dtype=object)
```

```
df["Contract Length"].unique()
```

```
array(['Monthly', 'Annual', 'Quarterly'], dtype=object)
```

```
Subscription_order = ['Basic', 'Standard', 'Premium']
Contract_order = ['Monthly', 'Annual', 'Quarterly']
oe1 = OrdinalEncoder(categories=[Subscription_order])

df["Subscription Type"] = oe1.fit_transform(df[["Subscription Type"]])
```

```
---------------------------------------------------------------------
-----
ValueError                                Traceback (most recent call
last)
~\AppData\Local\Temp\ipykernel_2952\1349088053.py in <module>
      3 oe1 = OrdinalEncoder(categories=[Subscription_order])
      4 oe2 = OrdinalEncoder(categories=[Contract_order])
----> 5 df["Subscription Type"] = oe1.fit_transform(df[["Subscription
Type"]])
      6 df["Contract Length"] = oe2.fit_transform(df[["Contract
Length"]])

c:\users\v.rajkumar\appdata\local\programs\python\python37\lib\site-
packages\sklearn\base.py in fit_transform(self, X, y, **fit_params)
    850         if y is None:
    851             # fit method of arity 1 (unsupervised
transformation)
--> 852             return self.fit(X, **fit_params).transform(X)
    853         else:
    854             # fit method of arity 2 (supervised
transformation)

c:\users\v.rajkumar\appdata\local\programs\python\python37\lib\site-
packages\sklearn\preprocessing\_encoders.py in fit(self, X, y)
    884
    885         # `_fit` will only raise an error when
`self.handle_unknown="error"`
--> 886         self._fit(X, handle_unknown=self.handle_unknown,
force_all_finite="allow-nan")
    887
    888         if self.handle_unknown == "use_encoded_value":

c:\users\v.rajkumar\appdata\local\programs\python\python37\lib\site-
packages\sklearn\preprocessing\_encoders.py in _fit(self, X,
handle_unknown, force_all_finite)
     94                 cats = _unique(Xi)
     95             else:
---> 96                 cats = np.array(self.categories[i],
dtype=Xi.dtype)
     97                 if Xi.dtype.kind not in "OUS":
     98                     sorted_cats = np.sort(cats)

ValueError: could not convert string to float: 'Basic'
```

```
oe2 = OrdinalEncoder(categories=[Contract_order])
df["Contract Length"] = oe2.fit_transform(df[["Contract Length"]])

df
```

```
       Age  Gender  Tenure  Usage Frequency  Support Calls  Payment
Delay  \
0       22       1      25               14              4
27
1       41       1      28               28              7
13
2       47       0      27               10              2
29
3       35       0       9               12              5
17
4       53       1      58               24              9
2
...    ...     ...     ...              ...            ...
...
64369   45       1      33               12              6
21
64370   37       0       6                1              5
22
64371   25       0      39               14              8
30
64372   50       1      18               19              7
22
64373   52       1      45               15              9
25

       Subscription Type  Contract Length  Total Spend  Last
Interaction  \
0                    0.0              0.0          598
9
1                    1.0              0.0          584
20
2                    2.0              1.0          757
21
3                    2.0              2.0          232
18
4                    1.0              1.0          533
18
...                  ...              ...          ...                .
..
64369                0.0              2.0          947
14
64370                1.0              1.0          923
9
64371                2.0              0.0          327
20
64372                1.0              0.0          540
13
64373                1.0              0.0          696
22
```

```
       Churn
0          1
1          0
2          0
3          0
4          0
...      ...
64369      1
64370      1
64371      1
64372      1
64373      1

[64374 rows x 11 columns]
```

```
X = df.iloc[:,:-1]
y = df.iloc[:,-1]
```

```
X.ndim
```

```
2
```

```
X_train,X_test,y_train,y_test =
train_test_split(X,y,test_size=0.2,random_state=42)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = LogisticRegression()
```

```
model.fit(X_train,y_train)
```

```
LogisticRegression()
```

```
y_pred = model.predict(X_test)
```

```
arrray_ytest = np.array(y_test)
array_ypred = np.array(y_pred)
```

```
comparision = np.concatenate((arrray_ytest.reshape(-
1,1),array_ypred.reshape(-1,1)),axis=1)
```

```
import sys
np.set_printoptions(threshold=sys.maxsize)
print(comparision)
```

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```
acc_score = accuracy_score(y_test,y_pred)
cm = confusion_matrix(y_test,y_pred)
recall = recall_score(y_test,y_pred)

print(f"Accuracy_score:{acc_score}")
print(f"Confusion Matrix:{cm}")
print(f"Recall_score:{recall}")
```

```
Accuracy_score:0.8320776699029127
Confusion Matrix:[[5660 1133]
 [1029 5053]]
Recall_score:0.830812232818152
```

```
### HyperParameter Tuning
from sklearn.model_selection import RandomizedSearchCV

params_grid = {"penalty":["l1", "l2"],
               "C":np.logspace(-4,4,20),
               'solver':["saga","liblinear","lbfgs"],
               'class_weight':[None,"balanced"],
                 }

grid = RandomizedSearchCV(
        estimator=model,
        param_distributions=params_grid,
        cv=3,
        scoring="accuracy",
        verbose=1,
        random_state=42,
        n_iter = 20
         )

grid.fit(X_train,y_train)
```

```
Fitting 3 folds for each of 20 candidates, totalling 60 fits

c:\users\v.rajkumar\appdata\local\programs\python\python37\lib\site-
packages\sklearn\model_selection\_validation.py:372: FitFailedWarning:

15 fits failed out of a total of 60.
The score on these train-test partitions for these parameters will be
```

```
set to nan.
If these failures are not expected, you can try to debug them by
setting error_score='raise'.

Below are more details about the failures:
--------------------------------------------------------------------
----------
15 fits failed with the following error:
Traceback (most recent call last):
  File "c:\users\v.rajkumar\appdata\local\programs\python\python37\
lib\site-packages\sklearn\model_selection\_validation.py", line 680,
in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "c:\users\v.rajkumar\appdata\local\programs\python\python37\
lib\site-packages\sklearn\linear_model\_logistic.py", line 1461, in
fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File "c:\users\v.rajkumar\appdata\local\programs\python\python37\
lib\site-packages\sklearn\linear_model\_logistic.py", line 449, in
_check_solver
    % (solver, penalty)
ValueError: Solver lbfgs supports only 'l2' or 'none' penalties, got
l1 penalty.

  warnings.warn(some_fits_failed_message, FitFailedWarning)
c:\users\v.rajkumar\appdata\local\programs\python\python37\lib\site-
packages\sklearn\model_selection\_search.py:972: UserWarning: One or
more of the test scores are non-finite: [0.82296749 0.50866973
0.82391898 0.8246957         nan 0.82473454
 0.82471512        nan 0.82424908 0.82388015 0.8242685         nan
 0.82391898 0.82477337 0.82473454        nan 0.82471512 0.82514229
        nan 0.82471512]
  category=UserWarning,

RandomizedSearchCV(cv=3, estimator=LogisticRegression(), n_iter=20,
                   param_distributions={'C': array([1.00000000e-04,
2.63665090e-04, 6.95192796e-04, 1.83298071e-03,
       4.83293024e-03, 1.27427499e-02, 3.35981829e-02, 8.85866790e-02,
       2.33572147e-01, 6.15848211e-01, 1.62377674e+00, 4.28133240e+00,
       1.12883789e+01, 2.97635144e+01, 7.84759970e+01, 2.06913808e+02,
       5.45559478e+02, 1.43844989e+03, 3.79269019e+03,
1.00000000e+04]),
                                        'class_weight': [None,
'balanced'],
                                        'penalty': ['l1', 'l2'],
                                        'solver': ['saga',
'liblinear',
                                                   'lbfgs']},
                   random_state=42, scoring='accuracy', verbose=1)
```

```python
print(f"Best_score:{grid.best_score_}")
print(f"Best_params:{grid.best_params_}")
```

```
Best_score:0.8251422937047451
Best_params:{'solver': 'saga', 'penalty': 'l2', 'class_weight': None,
'C': 0.00026366508987303583}
```